and phosphorus. They are excellent sources of B complex vitamins particularly, thiamine, folic acid and pantothenic acid. Like cereals, they do not contain any vitamin A or vitamin C, but germinated pulses contain vitamin C.

Pulses are considered as super food because they are:

- Good source of protein
- Rich in minerals and B vitamins
- Low in fat
- Gluten free
- Source of dietary fibre

2.10.2 Germination

Whole pulses are soaked overnight, water should be drained away and the seeds should be tied in a loosely woven cloth and hung. Water should be sprinkled twice or thrice in a day. In a day or two, germination takes place.



Fig 2.10: Germination

DO YOU KNOW...?

Pulses have been essential part of our diet for centuries. Agricultural production of pulses dates back to 7000-8000 BC.

ACTIVITY - 3

Germinate pulses and prepare any two products from them.

Advantages of germination

- Nutritive value is improved during sprouting. During sprouting, minerals like calcium, zinc and iron are released from bound form. Vitamin C is synthesized during germination.
- Sprouting decreases cooking time.
- Thickening power of starch is reduced due to conversion of starch to sugars.
- Germination improves taste and texture.
- Germinated pulses add variety to the diet.

2.10.3 Toxic constituents in pulses

Some pulses contain chemical constituents that have toxic properties.

1. **Trypsin inhibitors:** They are present in red gram, Bengal gram, cowpea, double bean, soyabean and dry peas. Trypsin inhibitors are proteins that inhibit the activity of trypsin in the gut and interfere with digestibility of dietary proteins and reduce their utilisation.
2. **Lathyrogens:** Lathyrism is a nervous disease that cripples man. It is known to result from an excessive consumption of the pulse kesari dhal (Lathyrissativus). The symptoms of lathyrism are muscular rigidity, weakness and paralysis of the leg muscle.

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